Problem Set-I (PH-2017), Relativity
Department of Physics

1. Tow lumps of clay, each of rest rest mass m, colliding head-on at $\frac{3}{5}c$. If they stick together then determine the mass of the composite lump.

2. A pion at rest decays into a muon (μ^-) and a neutrino (ν_μ). Find the momentum of the outgoing muon interms of the two masses m_π and m_μ assuming that the neutrino rest mass is zero.

3. A particle of rest mass m whose total energy is twice its rest energy collides with an identical particles at rest. If they stick together, what is the mass of the resulting composite particle ? What is its velocity ?

4. A neutral pion of rest mass M and momentum $P = \frac{3}{4}Mc$ decays into two photons (both massless). One of the photons is emitted in the same direction as the original pion and the other in the opposite direction. Find the relativistic energy of each photon.

5. A record turn table of radius R rotates at an angular velocity w. What is the ratio of the circumference to the diameter, in terms of w and R.

6. A muon is travelling through the laboratory at $\frac{3}{5}c$. How long does it lasts if its life time is 2×10^{-6} seconds in a rest frame.

7. Find the momentum and velocity of an electron having a kinetic energy $T = 10$ MeV. The rest energy of the electron is 0.512 MeV.

8. What is the length of a meter stick moving parallel to its length when its mass is 1.5 times its rest mass.

9. Show that the circle in a rest frame defined by $x^2 + y^2 = r^2$ appears to be an ellipse in a relativistic frame moving with a velocity v parallel to X-axis.

10. A circular lamina moves with its plane parallel to the X-Y plane of a reference frame S at rest. Assuming its motion to be along the X-axis, calculate the velocity at which its surface area would appear to be reduced to half to an observer in frame S.